

TRAINING REPORT

Extraction and Isolation of Bioactive Compounds from Medicinal and Aromatic Plants

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AT BIOTECH PARK LUCKNOW



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Thanking you,

**SUYASHI SINGH
RAJAT AWASTHI**

**BTECH BIOTECHNOLOGY
SHUATS
PRAYAGRAJ**

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OVERVIEW OF BIOTECH PARK

The state of Uttar Pradesh, a vital hub of scientific activities endowed with rich biological resources and rich biodiversity have the unique distinction of having a number of research institutions, which have expertise and capabilities in biotechnology.

In view of this, Lucknow was declared as the Biotechnology city of India during the 89th session of the Indian Science Congress on January 3, 2002. These institutions are providing a great impetus and support in the development of Biotech Park. The Park in turn provides incubator facilities to these institutions and entrepreneurs to perfect the facility, upscale the products and even produce small quantity for testing and quality control.



The Biotechnology Park set up on 8 acres of land provided the Department of Science and Technology, Government of Uttar Pradesh. The thrust areas identified for the initial phase of the park are: Health Care- Human and animal health care products including, immunodiagnosics, biosensors, vaccines, gene therapy, monoclonal antibodies, nutraceuticals and cosmeceuticals. Agriculture - improvement in the quality and yield of crops, horticulture and forest tree species, bio-pesticides and bio-fertilizers, processed food and quality enhancers; Environment Bio-remediation, safe disposal of waste; Industrial Application-Plants as bioreactors, enzymes, chemicals and polymers and energy Bio fuels and renewable energy sources.

SOPHISTICATED ANALYTICAL TESTING TECHNIQUES

The sophisticated analytical testing facility is located in an area 2280 sq. ft. The analytical Quality Assurance and control Laboratory is NABL accredited and meets ISO 9000 requirements. The facility has separate instrument, sample processing and microbiology laboratories. The state of art equipment includes HPLC, HPTLC, GC, and Atomic Absorption Spectrophotometer, Polari Spectrophotometer and other equipment. meter, Nino-Drop

BIOINFORMATICS CENTRE

A Bioinformatics centre has been setup to establish a close network with various institutions and provide information to industries regarding technological advancements. The bioinformatics centre organizes workshops sponsored by DBT every year. The major objective of bioinformatics centre is to showcase the R&D and the academic strength of Lucknow in the area of Biotechnology & Bioinformatics and act as noble point for industry -academia -U.P. Government.

The centre is working in close network with the neighbouring academic and R&D institutions to provide information to industries/ entrepreneurs/ farmers regarding technologies, facilities and expertise available with them in the area of Biotechnology

BIOFERTILIZER UNIT

The indiscriminate use of chemical fertilizer during past four decades. for increasing the agricultural produce has affected the soil fertility. The water retaining capacity and the contents of the nutrients like sulphur, boron, and zinc have also decreased in the soil. Considering this, the concept of organic farming in which the biofertilizers are. The use of bio-fertilizers improves the physical and chemical properties and fertility of the soil. The bio-fertilizers increased the humus content, porosity and water holding capacity and maintain the nutritional balances in the soil. The organic manures like compost, green manures, microbial inoculants such as blue green algae and Azolla, Rhizobium, Azotobacter, Phosphate solubilizing bacteria and Trichoderma are quite popular Bio-fertilizers, and their utility is well established.

TISSUE CULTURE UNIT

Tissue culture facility at Biotech Park, Lucknow is spread over 4071 sq. ft. area having capacity to raise and multiply Banana, Potato, Jatropha seedlings. The culture media used in the unit is completely biodegradable and non-toxic.



Advantages of propagation by Tissue culture

- The elimination of disease and the production of disease-free plantlets
- The rapid production of large number of genetically identical plantlets
- Introduction of new varieties and or genotypes
- Preservation of germplasm
- Production of haploid plants which can be used for plant breeding
- Production of plantlets from species in which plant development from seed is difficult
- Capacity
- 10000 to 100000 plants/batch and it can produce about million plants/annum.

DISTILLATION UNIT

Distillation unit is used for distillation of essential oils from a majority of the aromatic crops. i.e., *Mentha arvensis*, *Mentha piperita*, *Mentha cardiac*, *Geranium*, Lemongrass, Palma Rosa, Citronella, Basil and Vetiver.

Features

- Improved design
- All contact parts of stainless steel
- Oil yield more than the conventional units

- Low oil loss
- Utilizes spent Market/Agri waste as a fuel
- Less fuel consumption
- Easy loading/unloading through chain pulley system
- Environment friendly Capacity: 1000 kg fresh herbs/batch

EXTRACTION UNIT

The extraction unit comprises of solid liquid solvent extraction system with solvent extraction system recovery system for extraction of Photochemical/lead molecules from high value medicinal plants and multipurpose reaction cum hydrolysis and solvent recovery unit along with chromatography.

Oil fired steam boiler, evaporation capacity, 500kg/hr and 150psi steam pressure for basic heating requirements

- Heat air dryer, high-capacity hammer mill and vacuum oven for drying grinding/pulverizing of raw material/finished herbs/medicinal plants.
- Refrigerated brine chilling circulation unit for carrying out reactions and chilling of condenser water.
- Solid liquid extraction unit for consisting of two drug holders with agitators of 125-150 kg/batch capacity depending on plant bulk density with efficient solvent recovery system from extract and spent mare.
- Silica gel chromatography column for chromatography separation over silica gel with solvent recycling system, for enriching and isolation of the phytochemical in high.

HARDENING FACILITY

Hardening facility consists or polyhouses, Net houses and Glass house is a facility to protect and maintain the plants of temperate climate. These plants can also be multiplied in large scale during off season. The value crops like Geranium, patchouli, and pyrethrum could be saved during rainy season when the climate is hot and humid.

Advantages

- It is maintained under the guidance of trained technical staff
- Compartmentalized glass house for different conditions
- Provision for addition of specialized facilities like automatic temperature/ humidity/ mist control.

VERMICOMPOSTING UNIT

Vermicomposting is the fastest and effective way of recycling of organic waste with the help of earthworms to produce vermicompost. A huge quantity of distillation waste and other Agri-waste is generated at farm which is either burnt or rots in the field. It has been found that composting period is reduced in case distillation waste is used as Substrate by at least 20-35 days than vermicomposting of conventional wastes like wheat or rice straw, vegetable waste etc. Use of distillation waste primarily saves the pre-decomposition period required in the conventional wastes as the herbage after passing through high steam pressure during distillation completely breaks down facilitating the faster composing process. Pits- 16 pits (5.5m *0.8m each) Capacity-Vermicompost yield 350 -450kg vermicompost from 1.5 tons fresh herbs/ pits in 100 days

EQUIPMENT USED IN EXTRACTION UNIT

1- Multiple solvent extraction unit (leaching)

- Capacity 250-300 kg per batch with two extractors
- Material- pure stainless steel unit Facility for cold & hot extraction unit
- Extraction with agitator Application-Solvent & alcoholic extraction of medicinal herbs
- -Preparation water aqueous extract Extraction of valuable aroma chemical & spices
- Extraction of phytochemical

2- Liquid-liquid extraction column

- Capacity 20 hour continue counter current flow Downstream processing unit for purification/ liquid portioning extract
- High no of theoretical stage for maximum mass transfer

- Lesser floor area & less solvent requirement
- Application To selective portioning & Isolation of crude extract,
- phytochemical

3- Heavy duty GMP spray dryer

- Capacity 10 kg/hr water evaporation rate
- It built with rotating disc atomizer & two fluid nozzle systems
- Application. To make powder aqueous extract

4 Falling film evaporator Capacity 50 litre

Operate in short duration & low temp. condition Application Concentration of herbal extract

5- Oil fired steam boiler

- Capacity-500 kg/hr steam generation rate
- Fire tube type Operate in diesel oil
- Working pressure-150psig Automatic Moberly water level controller
- Fully automatic fuel jet firing on burner

6 Multiple reactions cum hydrolysis & solvent recovery unit along with.

chromatography column. Capacity 100lt/batch

60 kg silica gel with solvent recirculation pump

DISTILLATION AND ITS TYPE

Distillation refers to the selective boiling and subsequent condensation of a component in a liquid mixture. It is a separation technique that can be used to either increase the concentration of a particular component in the mixture or to obtain (almost) pure components of the mixture. is provided Distillation refers to the selective boiling and subsequent condensation of a component in a liquid mixture. It is a separation technique that can be used to either increase the concentration of a particular component in the mixture or to obtain (almost) pure components from the mixture. The process of distillation exploits the difference in the boiling points of the components in the liquid mixture by forcing one of them into a gaseous state.

It is important to note that distillation is not a chemical reaction but it can be considered as a physical separation process. An illustration describing the laboratory setup that is generally used to execute this process below.

Types of Distillation

Some important types of distillation include:

- Simple distillation
- Fractional distillation
- Steam distillation
- Vacuum distillation
- Air-sensitive vacuum distillation
- Short path distillation
- Zone distillation

Simple Distillation

Simple distillation involves heating the liquid mixture to the boiling point and immediately condensing the resulting vapor.

This method is only effective for mixtures wherein the boiling points of the liquids are considerably different (a minimum difference of 25°C). The purity of the distillate (the purified liquid) is governed by Raoul's law.

Fractional Distillation

Fractional distillation is often used to separate mixtures of liquids that have similar boiling points. It involves several vaporization-condensation steps (which takes place in a fractioning column). This process is also known as rectification. The apparatus required to perform a fractional distillation on a mixture is listed belonging **Round-bottom flask or distilling flask** A source of heat, which can be a fire or a hot bath. Receiving flask to collect the condensed vapor

Fractioning column

Thermometer to measure the temperature in the distilling flask
Condenser

Standard Glassware.

When heated, the liquid mixture is converted into vapor that rise into the fractioning column. The vapor now cool and condense on the walls of the condenser. The hot vapor emanating from the distilling flask now heat the condensed vapor, creating new vapor.

Many such vaporization-condensation cycles take place and the purity of the distillate improves with every cycle. An illustration depicting a fractional distillation setup is provided below.

Steam Distillation

Steam distillation is often used to separate heat-sensitive components in a mixture.

This is done by passing steam through the mixture (which is slightly heated) to vaporize some of it. The process establishes a high heat-transfer rate without the need for high temperatures.

. The resulting vapor is condensed to afford the required distillate. The process of steam distillation is used to obtain essential oils and herbal distillates from several aromatic flowers/herbs.

Vacuum Distillation

. Vacuum distillation is ideal for separating mixtures of liquids with very high boiling points. In order to boil these compounds, heating to high temperatures is an inefficient method. Therefore, the pressure of the surroundings is lowered instead.

The lowering of the pressure enables the component to boil at lower temperatures. Once the vapor pressure of the component is equal to the surrounding pressure, it is converted into a vapor.

These vapours are then condensed and collected as the distillate. The vacuum distillation method is also used to obtain high-purity samples of compounds that decompose at high temperatures.

Air-Sensitive Vacuum Distillation

For compounds that are sensitive to air and readily react with it, the vacuum distillation process is carried out but the vacuum must be replaced with an inert gas once the process is complete. Such a process is often referred to as air-sensitive vacuum distillation.

Short Path Distillation

Short path distillation is used to purify a small quantity of a compound that is unstable at high temperatures. This is done under lowered pressure levels and generally involves the distillate traveling a very small distance before being collected (hence the name 'short path'). The reduced distance travel by the distillate in this method also reduces the wastage along the walls of the apparatus.

Zone Distillation

The process of zone distillation involves the partial melting of a substance and the condensation of the resulting vapor to obtain a pure distillate. This is carried out in a long container with the help of a zone heater.

ESSENTIAL OIL

INTRODUCTION TO ESSENTIAL OIL

The term essential indicates that the oil carries distinctive scent essence of the plant, not that it is an especially important or fundamental substance. Essential oils do not as a group need to have any specific chemical properties in common, beyond conveying characteristic fragrances. They are not to be confused with essential fatty acids. Essential oils were generally extracted by distillation. Other processes include expression or solvent extraction. They are used in perfume and cosmetics, for flavouring food and drink, and for scenting incense and household cleaning products.

Various essential oils have been used medicinally at different periods in history. Medical applications proposed by those who sell medicinal oils range from skin treatments to remedies for cancer, and are often based on historical use of these oils for this purpose. Such claims are now subject to regulation in most countries, and have grown correspondingly vague, to stay within these regulations. Interest in essential oil has revived in recent decades, with the popularity of aromatherapy, a branch of alternative medicine which claims that the specific aromas carried by essential oils have curative effect. Oils are volatilized or diluted in carrier oil and used in massage, or burned as incense. The word "essential" refers to the fact that each contains the imprint of the plant's specific qualities, which act as a protection of the plant itself. The volatile molecules of the essential oil communicate with the plant's environment, and with mankind, through aroma and vibratory rate.

Essential oil chemistry

Essential oils are made up of many chemical constituents. No two oils are alike in their structure or their effects. Each of these can be broken down into numerous smaller units. Take terpenes, for example; it includes monoterpenes, sesquiterpenes, sesquiterpene lactones, Di-terpenes, etc. Listing them all is beyond the scope of this brief overview.

Each constituent has its own action, or effects. For example, the ketones found in lavender, hyssop and patchouli, stimulate cell regeneration. Whereas, phenols, found in oregano and thyme oil, are highly antimicrobial, because the wide variety of antiseptic compounds in essential oils makes the mutation of microorganisms extremely difficult.

EXPERIMENT NO. 1

AIM: To obtain essential oil from fresh lemongrass without chopping

MATERIAL REQUIRED

Clevenger apparatus, electronic weighing machine distilled water

PROCEDURE

1. 250 gm Lemon grass leaves were taken in the round bottle flask of the Clevenger apparatus
2. The round bottom flask was filled with water up to heating filament.
3. Continue water supply in condenser.
4. The Clevenger apparatus was switched on.
5. Initially 50 degrees Celsius was kept. When water starts boiling then temperature is gradually increase to 60 degrees Celsius.
7. This process of extraction of essential oils was maintained for 3-4 hours.
8. After some time of oil was noted and observed in the scale of Clevenger apparatus, for the duration every 30 minutes, reading the oil was noted and this process continued for 3-4 hour.
9. After 3-4 hours two layers were observed in the Clevenger scale, the lower layer was of water and upper layer contained extracted essential oil.
10. When in Clevenger apparatus, oil reading is constant with increased temperature.
11. The two layers were carefully separated with the help of Clevenger knob.
12. Oil and water were collected in two separate small viols.
13. Oil was carefully weighed and percentage yield was calculated

OBSERVATION

SNO.	TIME	TEMPERATURE	OIL ML
1	12: 20pm	50	00
2	12 50	50	00
3	01 20	50	2.4
4	01 50	50	3
5	02 20	50	3
6	02 50	60	3.6

RESULT 3.6 ml essential oil is obtained successfully from 250gm lemongrass.



EXPRIMENT NO:-2

AIM To obtain essential oil from 1-day dry lemongrass with chopping

MATERIAL REQUIRED distilled water, leaves at 100gm

OBSERVATION

SNO.	TIME	TEMPERATURE	OIL ML
1	11: 40am	50	00
2	12 10	50	00
3	12 40	50	1ml
4	01 10	50	2.8
5	01 40	50	2.9
6	02 10	50	3
7	02 40	50	3.3
8	03 10	50	3.3
9	03 40	60	3.5

RESULT 3.5 ml essential oil obtained from 1day dried lemongrass



EXPERIMENT 3

AIM To obtained essential oil from 2 days dried lemongrass.

MATERIAL REQUIRED distilled water, lemongrass, Clevenger apparatus

OBSERVATION

SNO.	TIME	TEMPERATURE	OIL ML
1	11 10am	50	00
2	11 40	50	00
3	12 10	50	1ml
4	12 40	50	2ml
5	01 10	50	2.5
6	01 40	50	2.6
7	02 10	50	2.6
8	02 40	60	3.1

RESULT 3.1 ML essential oil from 2 days dried lemongrass leaves .



EXPERIMENT 4

AIM TO obtain essential oil from 3 days dried lemongrass leaves

MATERIAL REQUIRED distilled water, Cleavinger apparatus, lemongrass wt150gm

OBSERVATION

SNO.	TIME	TEMPERATURE	OIL ML
1	11: 05am	50	00
2	11 35	50	00
3	12 05	50	2.4
4	12 35	50	2.6
5	01 05	50	2.8
6	01 35	50	3
7	02 05	50	3
8	02 35	60	3.4

RESULT 3.4ml essential oil obtained from 3 days dried lemongrass leaves.

EXPERIMENT 5

AIM To obtain essential oil from 4 days dried lemongrass

MATERIAL REQUIRED distilled water, lemongrass 100gm

OBSERVATION

SNO.	TIME	TEMPERATURE	OIL ML
1	12 00pm	50	00
2	12 30	50	00
3	01 00	50	1.3
4	01 30	50	2
5	02 00	50	2.8
6	02 30	50	3.5
7	03 00	50	3.5
8	03 30	60	3.7

RESULT 3.7ml essential oil obtained from 4-day dried lemongrass.

EXPERIMENT 6

AIM To obtain essential oil from 5-day dried lemongrass leaves

MATERIAL REQUIRED distilled water, Clevenger apparatus, lemongrass leaves at 150gm.

OBSERVATION

SNO.	TIME	TEMPERATURE	OIL ML
1	11 25 am	50	00
2	11 55	50	00
3	12 25	50	0.5
4	12 55	50	2.5
5	01 25	50	2.8
6	01 55	50	3
7	02 25	50	3.5
8	02 55	50	3.5
9	03 25	60	3.8

RESULT 3.8ml essential oil obtained from 5 days dried lemongrass.

EXPERIMENT 7

Aim: To determine the moisture content of oil of lemon grass (*Cymbopogon* sp)

Principle:- Moisture content is the quantity of water contained in a material such as soil, ceramics, rock or wood on a volumetric or gravimetric basis. The property is used in a wide range of scientific and technical areas and is expressed as a ratio, which can range from 0 to the value of the material's porosity at saturation.

Apparatus/Requirement

Freshly harvested herbs of lemon grass

Balance

Drying oven with thermostat and temperature indicator

Drying tray.

Procedure :

- Weight 1000 gm freshly harvested herb
- Take weight of an empty beaker.
- Switched on the drying oven and set the thermostat at 50 degrees Celsius. Take 1 ml of sample in the beaker and record the weight.
- Spread the *Cymbopogon citratus* on the thermostat at 50 degrees Celsius. Keep it in oven for 15 minutes at 50-degree C.
- Switched off the oven after 6 or more hours and allow to cool for one hour.
- Keep it in the desiccator and allow it to cool to room temperature.
- Record the weight of dry herb.
- Keep it in the oven for 30 minutes again and cool to the room temperature.
- Repeat the procedure till the weight comes to be constant.

Formula :

Moisture content = $\frac{\text{Initial wt. of sample} - \text{final wt. of sample}}{\text{Initial wt. of sample}} \times 100$

Initial wt. of sample

Observation :

Initial wt. of sample = 1000 gm

Calculation

Final wt. of sample = 350 gm

Moisture content $1000 - 350 \times 100$

1000

650 X 100

1000

= 65%

EXPERIMENT 8

Objective: To determine bulk density of lemon grass.

Principle: The bulk density of any herb indicates the amount / mass of the herb that will occupy a certain volume of the vessel it is kept in. It is a useful tool / parameter used for determining/calculating the capacity and designing a distillation unit tank.

Apparatus / Requirements Freshly harvested herbs of lemon grass . Balance Measuring flask/beaker

Procedure

- . Measure the tare weight of empty flask/beaker
- Fill the plants material uniformly up to the marks
- Weight the beaker +plant materials
- Subtract the tare weight of beaker
- Calculate the bulk density of the plants material as follows

Bulk Density: weight of the plant material (gm)/volume of beaker (in cylinder= 254 gm. = 1 litre

Wt. of cylinder = 254 gm

Wt. Of cylinder + sample = 476g

Wt. Of lemongrass sample = 222g

Bulk density

= $222 / 1000 = 0.22\text{g/ml}$



EXPERIMENT 9

AIM TO OBTAIN ESSENTIAL OIL FROM FRESH EUCALYPTUS LEAVES 300GM

MATERIAL REQUIRED distilled water, leaves 300gm, Clevenger apparatus.

OBSERVATION

SNO.	TIME	TEMPERATURE	OIL ML
1	11: 35 am	50	00
2	12 05	50	00
3	12 35	50	0.5
4	01 05	50	1.4
5	01 35	50	1.8
6	02 05	50	2
7	02 35	50	2
8	03 05	60	2.1

RESULT 2.1 ml of essential oil obtained from fresh eucalyptus leaves



EXPERIMENT 10

AIM To OBTAIN essential OIL from eucalyptus leaves after 3 days dry

MATERIAL REQUIRED distilled water, Clevenger apparatus, eucalyptus leaves original wt. 600gm present wt. 200gm

OBSERVATION

SNO	TIME	TEMPERATURES	OILML
1	12 00pm	50	00
2	12 30	50	00
3	01 00	50	0.1
4	01 30	50	1.4
5	02 00	50	2.5
6	02 30	50	3.0
7	03 00	50	3.0
8	03 30	60	3.4

RESULT 3.4 ml of essential oil obtained from 3 days dry eucalyptus leaves.

EXPERIMENT 11

AIM To obtain essential OIL from fresh MENTHA leaves along with stems wt. 250 gm

THEORY The species that make up the genus *Mentha* are widely distributed and can be found in many environments. Most grow best in wet environments and moist soils. Mints will grow 10–120 cm (4–48 inches) tall and can spread over an indeterminate area. Due to their tendency to spread unchecked, some mints are considered invasive.

MATERIAL REQUIRED distilled water, weighing machine ,Cleavinger apparatus

OBSERVATION

SNO	TIME	TEMPERATURES	OIL ML
1	11 25am	50	00
2	11 55	50	00
3	12 25	50	2.8
4	12 55	50	3.8
5	01 25	50	5
6	01 55	50	5.8
7	02 25	50	5.8
8	02 55	60	6.1

RESULT 6.1 ml of essential oil obtained from fresh MENTHA



EXTRACTION 12

AIM To obtain moringa extract from fresh moringa leaves

THEORY Moringa is rich in amino acids which are building blocks of cells; therefore, this helps in rejuvenation of hair thereby, reduces alopecia. **It activates the dormant hair follicles thereby enhancing hair growth on the bald areas.** It is a rich source of Biotin.

MATERIAL REQUIRED Soxhlet apparatus, fresh wt. 64gm dry wt. 22gm ,solvent.

OBSERVATION



RESULT extract is successfully obtained

EXPERIMENT 13

AIM To obtain neem extract from dry neem powder.

THEORY Neem is a natural herb that comes from the neem tree, other names for which include Azadirachta indica and Indian lilac. The extract comes from the seeds of the tree and has many different traditional uses. Neem is known for its pesticidal and insecticidal properties, but people also use it in hair and dental products.

MATERIAL REQUIRED dry neem powder, solvent , Soxhlet apparatus.

OBSERVATION

Day	Process time	Temp
1	72hrs/ 01 00pm	30
2	“	30
3	“	40

RESULT



EXPERIMENT 14

Aim: Preparation of Aloe gel from Aloe vera leaves

Apparatus/Requirements Clevenger apparatus. Aloe vera leaves Fresh filter water Beaker Carbopol gelling agent

Procedure: -

Take fresh healthy Aloe vera leaves.

Cut from both sides press the leaves.

Collect the gel in a fresh beaker.

Grind the gel.

Evaporate the gel and then cool down.

Add 6 gm Carbopol and then leave 6 to 8 hours(1-2)

Check for acidity and basicity

Result: - Freshly prepared aloe vera gel is collected.



CONCLUSION

LEMONGRASS OIL-

In this paper, extraction of essential oil from lemongrass which has high essential oil content, which will be used for used for perfume formulation. Analyses will be carried out to determine the various oil yields using different extraction methods and the formulation of perfume with the essential oil will be produced. All these methods of extraction are special type of separation process used for heat sensitive materials like essential oils, resins, hydrocarbons, etc, which are insoluble in water and may decompose at their boiling point. The temperature of the steam must be high enough to vaporize the essential oil present, yet not destroy or burns the essential oils.

EUCALYPTUS

It is a fast-growing evergreen tree native to Australia. As an ingredient in many products, it is used to reduce symptoms of coughs, colds, and congestion. It also features in creams and ointments aimed at relieving muscle and joint pain.

The oil that comes from the eucalyptus tree is used as an antiseptic, a perfume, as an ingredient in cosmetics, as a flavouring, in dental preparations, and in industrial solvents.

Chinese, Indian Ayurvedic, Greek, and other European styles of medicine have incorporated it into the treatment of a range of conditions for thousands of years.

There are over 400 different species of eucalyptus. Eucalyptus globulus, also known as Blue Gum, is the main source of eucalyptus oil used globally.

Leaves are steam distilled to extract the oil, which is a colourless liquid with a strong, sweet, woody scent. It contains 1,8-cineole, also known as eucalyptol.

The leaves also contain flavonoids and tannins; flavonoids are plant-based antioxidants, and tannins may help to reduce inflammation.

ALOE VERA GEL-

Aloe vera gel subjected to HHP treatments at 150, 250, 350, 450, and 550 MPa for 5 minutes, was affected in terms of quality during processing and storage. Antioxidant capacity, total phenolic content, colour, firmness, rehydration ratio, and water holding capacity measured after 60 days storage at 4 °C showed significant changes. Total phenolic content showed an increase as pressure level increased, which is associated with cell disruption and release of these compounds. The best colour retention of samples could be observed between the operating pressures of 150 and 250 MPa. Modifications in cell wall structure due to HHP also resulted in a decrease in the gel firmness as well as in its rehydration capacity meaning loss in the initial ability to retain water. These results showed that this innovative HHP technology has a potential use in the preservation of aloe vera gel quality, which would be highly beneficial for the production of a high-quality ingredient.

MORINGA , NEEM , MENTHA AND MAMY MORE.....

Thank you